

Statistical Methods IV Project

Non-Stationary Time Series Analysis of Temperature Data in India



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April 2026

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1 INTRODUCTION

Understanding long-term variations in temperature is central to studying climate dynamics and their broader societal implications. In recent years, there has been increasing interest in analyzing temperature data not merely for descriptive purposes, but to uncover underlying structural patterns such as trends, seasonal effects, and stochastic fluctuations. Such data typically exhibit strong non-stationary behavior, making them particularly suitable for advanced time series modeling.

In this project, we analyze the dataset titled “*Global Land Temperatures By Country*”, focusing specifically on temperature observations for India. This dataset contains historical records of average land temperatures aggregated at the country level over time. The richness of the dataset lies in its long temporal span, which allows for the study of both short-term seasonal variations and long-term climatic trends.

1.1 Objectives and Research Questions

The primary objective of this study is to analyze the temporal behavior of temperature data in India using non-stationary time series techniques. In particular, we aim to address the following questions:

- Does the temperature time series exhibit clear evidence of non-stationarity?
- What are the dominant components (trend, seasonality, irregular fluctuations) present in the data?
- How can we appropriately transform the data to achieve stationarity?
- What insights can be drawn about the underlying temperature dynamics over time?

These questions guide both the exploratory analysis and the subsequent modeling approach.

1.2 Statistical Methodology

The analysis in this project proceeds in a structured manner, beginning with data pre-processing and followed by exploratory and inferential techniques. The key steps are outlined below:

1. **Data Cleaning:** The dataset is first loaded and preprocessed by removing missing values to ensure consistency and reliability of the analysis.
2. **Exploratory Data Analysis:** We perform an initial summary statistics of the data and use plots to check for trend and seasonality.
3. **Testing for Stationarity:** We use the KPSS test to confirm non-stationarity
4. **Data Splitting and Transformation:** We split the data into training(80% of the data) and test(remaining 20%).
5. **Model Identification and Fitting:** Based on the autocorrelation function (ACF) and partial autocorrelation function (PACF) plots, we identify and fit a suitable SARIMA model and later use Ljung-Box test to verify model accuracy
6. **Forecasting and Validation:** Finally, we generate forecasts using the fitted model and compare the predicted values with the actual observations from the test dataset.

These methodologies are explained further in details in the subsequent sections.

2 DATA DESCRIPTION

2.1 Source

The dataset used in this study is titled “*Global Land Temperatures By Country*”, and is publicly available on Kaggle.

The dataset contains monthly records of land temperatures for various countries starting from the year 1743. In its raw form, the dataset consists of 577,462 observations and 4 variables, namely **Country**, **Date**, **Average Temperature**, **Average Temperature Uncertainty**.

For the purpose of this analysis, we restrict our attention to observations corresponding to **India** and further limit the time period to the years **1880 to 2013**. After this filtering, the dataset reduces to 2613 observations while retaining the same four variables.

2.2 Important Variables

Although the original dataset contains four variables, our analysis focuses primarily on two key variables:

- **Date:** This variable records the time of observation. It is treated as a time index and plays a crucial role in constructing the time series.
- **Average Temperature:** This represents the average land temperature (in degrees Celsius) for the given country and time period. This is the primary variable of interest in our study.

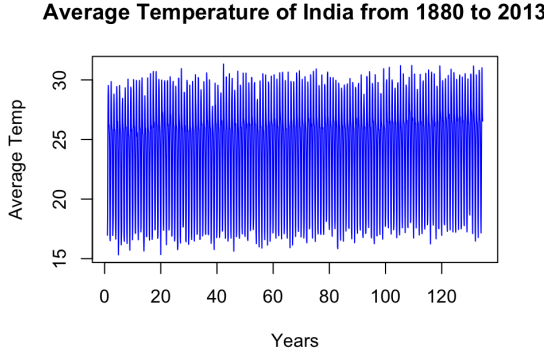
The data is subsequently converted into a **time series object** with a frequency of 12, corresponding to monthly observations within each year. This allows us to explicitly capture and analyze recurring seasonal patterns on an annual scale.

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
15.32	20.13	25.46	24.09	27.23	31.33

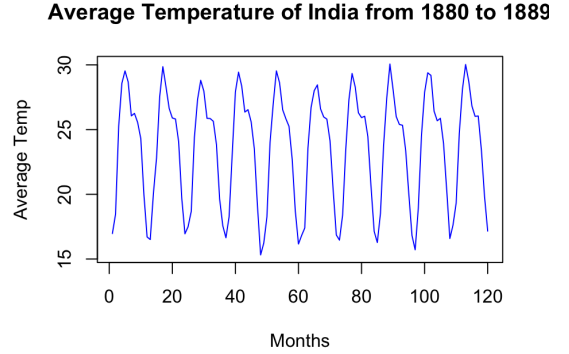
Table 1: Summary Statistics of *Temperature Data*

3 EXPLORATORY DATA ANALYSIS

We begin our analysis by visually examining the behavior of the temperature time series. A plot of the monthly average temperature for India over the considered time period is presented below.



(a) Average Temperature from 1880-2013



(b) Average Temperature from 1880-1889

From the time series plot, we make the following observations:

- We can see a clear recurring annual pattern with crests and troughs corresponding to each year.
- From the left plot, we can see a faint trend indicating that the mean average temperature is not constant over the years but increases gradually over the years.

Thus, the visual inspection of the data strongly suggests that the temperature series is non-stationary.

3.1 KPSS Test for Stationarity

To formally assess stationarity, we employ the KPSS (Kwiatkowski–Phillips–Schmidt–Shin) test. The key features of the test are as follows:

- **Null Hypothesis (H_0):** The time series is stationary (level or trend stationary).
- **Alternative Hypothesis (H_1):** The time series is non-stationary, typically due to the presence of a unit root or random walk component.

Test Statistic: Let $\hat{\eta}_t$ denote the residuals obtained after regressing x_t on an intercept (and possibly a time trend). Then the KPSS statistic is given by

$$\text{KPSS} = \frac{1}{n^2 \hat{\sigma}^2} \sum_{t=1}^n \left(\sum_{i=1}^t \hat{\eta}_i \right)^2,$$

where $\hat{\sigma}^2$ is a consistent estimator of the long-run variance of $\{\eta_t\}$.

Asymptotic Distribution: Under the null hypothesis, the statistic converges in distribution as

$$\text{KPSS} \xrightarrow{d} \int_0^1 V^2(x) dx,$$

where $V(x)$ is a **second-level Brownian bridge**.

Result and Interpretation: Implication:

Test	H_0	p -value	Conclusion
KPSS	Stationary	0.01	Stationary

Table 2: Result

Hence, both graphical analysis and the KPSS test confirm that the data is non-stationary, motivating the need for appropriate transformations.

3.2 Seasonal Mann–Kendall Test

The Seasonal Mann–Kendall (SMK) test detects a monotonic trend(H_1) in seasonal time series by applying the Mann–Kendall test within each season.

Let x_{ij} denote observation in season $i = 1, \dots, m$ and year $j = 1, \dots, n_i$.

Statistic

$$S_i = \sum_{k=1}^{n_i-1} \sum_{j=k+1}^{n_i} \text{sgn}(x_{ij} - x_{ik}), \quad S' = \sum_{i=1}^m S_i.$$

$$\text{Var}(S') = \sum_{i=1}^m \text{Var}(S_i), \quad \text{Var}(S_i) = \frac{1}{18} \left[n_i(n_i - 1)(2n_i + 5) - \sum t_{ip}(t_{ip} - 1)(2t_{ip} + 5) \right].$$

$$Z_{SMK} = \begin{cases} \frac{S' - 1}{\sqrt{\text{Var}(S')}}, & S' > 0, \\ 0, & S' = 0, \\ \frac{S' + 1}{\sqrt{\text{Var}(S')}}, & S' < 0. \end{cases}$$

Test	H_0	p -value	Conclusion
Seasonal Mann Kendall	Constant Trend	1.15×10^{-72}	Trend exists

Table 3: Result for seasonal mann-kendall test

3.3 Train–Test Split

We divide the time series into an 80:20 split, where the first 80% of observations form the training set and the remaining 20% are reserved for testing and forecasting. The split is performed chronologically and not randomly, preserving the temporal order of the data.

After partitioning the data, we retain 1283 months for training and the rest 321 for testing.

4 TRANSFORMATION TOWARDS STATIONARITY

4.1 Step 1: Regular Differencing to Remove Trend

To eliminate the underlying trend component, we first apply a first-order differencing to the training time series. The transformed series is defined as:

$$\nabla y_t = y_t - y_{t-1} \tag{1}$$

We again employ the SMK test to check if any trend still remains. A p -value of 0.49 suggests we fail to reject the null, indicating we have successfully removed the trend.

Thus we can say that there exists a linear trend in the data confirming our visual analysis.

4.2 Step 2: Seasonal Differencing to Remove Seasonality

Although the trend component is removed, the series may still exhibit seasonal patterns due to its monthly nature. To eliminate seasonality, we apply a seasonal differencing with lag 12. Applying this transformation to the previously differenced series, we obtain:

$$\nabla_{12}\nabla y_t = (y_t - y_{t-1}) - (y_{t-12} - y_{t-13}) \quad (2)$$

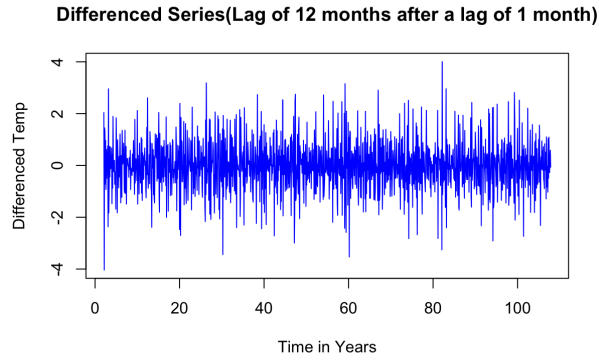


Figure 2: Differenced Data

As evident from the plot, we have removed trend as well as any seasonality.

4.3 Augmented Dickey–Fuller (ADF) Test

The Augmented Dickey–Fuller (ADF) test is a statistical test used to check for the presence of a unit root in a time series, and hence to assess whether the series is non-stationary.

The test is based on the regression:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^k \delta_i \Delta y_{t-i} + \varepsilon_t, \quad (3)$$

where $\alpha, \beta, \gamma, \delta_i$ are the regression coefficients for all i and $\Delta y_t = y_t - y_{t-1}$ and the lagged difference terms are included to account for higher-order autocorrelation.

The hypotheses for the test are:

- H_0 : $\gamma = 0$ (the series has a unit root, i.e., it is non-stationary),
- H_1 : $\gamma < 0$ (the series is stationary).

The test statistic is the t -statistic corresponding to the estimate of γ , given by:

$$\tau = \frac{\hat{\gamma}}{\text{SE}(\hat{\gamma})}. \quad (4)$$

Under the null hypothesis, this statistic does not follow the standard t -distribution. Instead, it converges to a non-standard distribution known as the Dickey–Fuller distribution, whose critical values are obtained through simulation.

A sufficiently small value of the test statistic (more negative than the critical value) leads to rejection of the null hypothesis, indicating that the series is stationary.

4.4 Confirmation of Stationarity

To formally confirm that the transformed series is stationary, we apply both the Augmented Dickey–Fuller (ADF) test and the KPSS test.

Test	H_0	p -value	Conclusion
ADF	Unit Root	0.01	Stationary
KPSS	Stationary	0.1	Stationary

Table 4: Result of ADF and KPSS Test

The agreement between these two tests confirms that the series has been successfully transformed into a stationary process. This justifies the use of ARMA-type models for subsequent analysis and forecasting.

5 MODEL IDENTIFICATION

5.1 Interpretation of ACF and PACF of the Transformed Series

The ACF and PACF are plotted up to lag 120 for the fully transformed (stationary) series. The following standard identification rules are used to guide model selection:

Table: ACF/PACF pattern guide for SARIMA order identification

Pattern	Non-seasonal order	Seasonal order
ACF cuts off at lag q , PACF tails off	MA(q)	—
PACF cuts off at lag p , ACF tails off	AR(p)	—
Both tail off	ARMA(p, q)	—
Spike at lag 12 in ACF	—	SMA($Q = 1$)
Spike at lag 12 in PACF	—	SAR($P = 1$)

- From the observed ACF and PACF plots, it is evident that the series exhibits a clear seasonal MA(1) component, as indicated by a significant spike at the seasonal lag.
- Further, the ACF shows a sharp cut-off after lag 1 or 2, while the PACF decays gradually. This pattern suggests that the underlying non-seasonal component can be reasonably modeled using an MA(1) or MA(2) process.

5.2 Candidate Models

Based on the interpretation of the ACF and PACF plots of the transformed series, we consider the following two candidate models:

$$\text{SARIMA}(0, 1, 1)(0, 1, 1)_{12} \quad \text{and} \quad \text{SARIMA}(0, 1, 2)(0, 1, 1)_{12}$$

5.3 Benchmark and Model Selection

To determine the more appropriate model, we fit both candidate models and compare their performance using the Root Mean Square Error (RMSE) and the Ljung–Box test for residual independence.

Model 1: SARIMA(0, 1, 1)(0, 1, 1)₁₂

We fitted the original undifferenced data to this model and calculated the RMSE and performed a Box-Ljung test.

$\chi^2 = 62.13$	df = 12	$p\text{-value} = 9.205 * 10^{-9}$
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Table 5: Box-Ljung test

Model 2: SARIMA(0, 1, 2)(0, 1, 1)₁₂

We perform the same on this model.

We also compute the RMSE for both models:

$\chi^2 = 13.431$	$df = 12$	$p\text{-value} = 0.3385$
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Table 6: Box-Ljung test

SARIMA(0, 1, 1)(0, 1, 1) ₁₂	SARIMA(0, 1, 2)(0, 1, 1) ₁₂
0.6008247	0.5863501

Table 7: RMSE of Residuals

The Ljung–Box test for Model 1 yields a very small p-value, indicating significant autocorrelation in the residuals and hence model inadequacy. In contrast, Model 2 produces a large p-value, suggesting that the residuals behave like white noise.

Furthermore, Model 2 achieves a lower RMSE compared to Model 1. Therefore, based on both residual diagnostics and predictive accuracy, we select:

$$\mathbf{SARIMA}(0, 1, 2)(0, 1, 1)_{12}$$

as the final model.

5.4 Final Model Representation

The selected model can be written explicitly as:

$$(1 - B)(1 - B^{12})y_t = (1 + \theta_1 B + \theta_2 B^2)(1 + \Theta_1 B^{12})\epsilon_t, \quad (5)$$

where B denotes the backshift operator, θ_1, θ_2 are the non-seasonal MA parameters, Θ_1 is the seasonal MA parameter, and $\{\epsilon_t\}$ is a white noise process.

ma1 (θ_1)	ma2 (θ_2)	sma1 (Θ_1)
-0.6630868	-0.2316589	-0.9700004

Table 8: Parameter Estimates using the *arima* function

6 FORECASTING

Having selected and fitted the final model, we proceed to generate forecasts for the upcoming 321 months from Dec 1986 to Aug 2013.

6.1 Residual Analysis

To assess the adequacy of the fitted model, we examine the residuals obtained from the model. A well-fitted time series model should produce residuals that behave like white

noise.

As shown before, our chosen model fails the Ljung–Box test indicating that the residuals are now white noise. Moreover, an ACF plot of the residuals also cut off after 0 which confirm that the model chosen is good.

6.2 Accuracy Metrics

To evaluate the performance of the fitted model, we consider the Root Mean Square Error (RMSE), defined as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n e_t^2}, \quad (6)$$

where e_t denotes the residual at time t .

A lower value of RMSE indicates a better fit to the observed data. This metric is particularly useful for comparing competing models and assessing overall prediction accuracy.

6.3 Prediction Interval Coverage

Prediction interval (PI) coverage measures the proportion of actual observations that fall within the constructed prediction intervals. It serves as an validation of the model whether it performs well outside the training data too.

In a SARIMA model, we assume the errors to have a normal distribution. Thus to build a 95% prediction interval, the coverage is defined as:

$$\text{Coverage} = \frac{1}{n} \sum_{t=1}^n \mathbf{1}\{y_t \in [L_t, U_t]\}, \quad (7)$$

where $[L_t, U_t]$ denotes the prediction interval at time t , and $\mathbf{1}\{\cdot\}$ is the indicator function.

$$L_t = \hat{y}_t - 1.96 * \hat{\sigma}$$

$$U_t = \hat{y}_t + 1.96 * \hat{\sigma}$$

where $\hat{y}_t$ are the predicted values and $\hat{\sigma}$ is the estimated standard deviation.

We obtain a coverage of 100% indicating that the model is not overfitted and captures future data well. **(We will talk about this in the Limitations section)**

7 RESULTS SUMMARY

Table: Summary of analysis pipeline and key findings

Stage	Decision / Finding
Data selection	India temperature data (1990–2013), 2613 observations after filtering
Data cleaning	Missing values removed; consistent time indexing ensured
Time series construction	Converted to monthly time series with frequency 12
EDA	Clear presence of seasonality and trend observed
KPSS (raw data)	Rejects stationarity $\Rightarrow$ transformation required
Trend removal	First-order differencing ($d = 1$) applied; trend eliminated (confirmed via Seasonal Mann–Kendall test)
Seasonality removal	Seasonal differencing at lag 12 ($D = 1$) removes periodic component
Stationarity check	ADF rejects unit root, KPSS fails to reject stationarity $\Rightarrow$ series is stationary
ACF/PACF interpretation	MA-type behavior with seasonal MA(1) component identified
Candidate models	SARIMA(0, 1, 1)(0, 1, 1) ₁₂ SARIMA(0, 1, 2)(0, 1, 1) ₁₂
Model selection	Based on RMSE and Ljung–Box test
Final model	SARIMA(0, 1, 2)(0, 1, 1) ₁₂
Ljung–Box test	$p > 0.05$ for final model $\Rightarrow$ residuals resemble white noise
RMSE	Lower RMSE achieved for selected model
Forecast horizon	321 months ahead
Prediction interval coverage	100% of actual values lie within confidence bands

8 LIMITATIONS

- In a SARIMA model, the errors are assumed to be Gaussian which may not be so for our data here. Infact, if we assume the errors to be normally distributed, we obtain a coverage of 100% which is highly unlikely. It maybe so that the errors

follow some other distribution whose coverage interval is smaller than that of normal distribution.

- Since global warming happens very slowly over the years, unless we take a large number of datapoints, we may not be able to capture the trend in the data.
- Average temperature of a country can depend on various other factors such as humidity, elevation, wind, etc. It will be a nice study if we can study the dependence structure of temperature on these factors which we cannot do due to limitation of data.
- It will be better if we could analyse daily records of temperature because a daily forecast of temperature would be useful to the public to plan ahead instead of a monthly data.

9 INFERENCE

- The temperature time series for India exhibits a strong seasonal pattern, with a clear periodicity of 12 months, indicating that corresponding months across years show similar temperature behavior.
- The presence of a gradual upward trend in the series suggests a slow but persistent increase in average temperature over time, providing statistical evidence consistent with global warming.
- Through thorough statistical techniques, we have been able to model this trend and seasonality using a $SARIMA(0, 1, 2)(0, 1, 1)_{12}$ model which also provides accurate forecasts.

10 ACKNOWLEDGEMENTS

We express our sincere gratitude to the creators and maintainers of the dataset “*Global Land Temperatures By Country*”, which is publicly available on Kaggle. The dataset provides valuable historical temperature records and serves as the foundation for the present analysis.

The dataset was accessed via the Kaggle repository: <https://www.kaggle.com/datasets/vijayvvenkitesh/global-land-temperatures-by-country>. We acknowl-

edge the efforts of the contributors in compiling, cleaning, and making this data publicly accessible for research and analysis.

We also acknowledge the use of the statistical programming language R (version ≥ 4.3) for all data processing, analysis, and modeling tasks. In particular, we have utilized packages such as `forecast`, `tseries`, `trend`, and `Kendall` for implementing various statistical procedures and time series models.

Finally, we would like to thank our instructors for their guidance and for providing the opportunity to explore non-stationary time series models through this project.

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12 R CODE

```
#Loading Necessary Packages
library(readr)
library(forecast)
library(Kendall)
library(tseries)
library(trend)

#Loading metadata and taking avg temp of India from 1880 to
  August 2013
GlobalLandTemperaturesByCountry <- read_csv("Datasets/
  GlobalLandTemperaturesByCountry.csv")
```

```

df<-data.frame(GlobalLandTemperaturesByCountry[which(
  GlobalLandTemperaturesByCountry$Country=="India"),])
temperatureData <- df[1009:2613,]
temperatureData <- na.omit(temperatureData$AverageTemperature)

#Let us perform a quick summary statistic of the data
summary(temperatureData)

#We are converting it into a time series with frequency 12 (
  yearly cycles)
avg_temp<-ts(temperatureData, frequency = 12)
plot(avg_temp, xlab= "Years", ylab = "Average Temp", main = "
  Average Temperature of India from 1880 to 2013", col="blue",
  lwd=1)

#For easy visualisation
plot(avg_temp[1:120], xlab = "Months", ylab = "Average Temp",
  main = "Average Temperature of India from 1880 to 1889", col="
  blue", lwd=1, type = "l")

#As evident from the plots, they show a clear seasonal pattern
#Performing a KPSS test to check if its stationary or not
s<-kpss.test(avg_temp)
ifelse ((s$p.value <0.05), "Non-Stationary", "Stationary")
#Judging by the p-value it is a non stationary process
#We will now employ a Seasonal Mann-Kendall trend test to check
  for any trend in this seasonal data
smk_test<-smk.test(avg_temp)
ifelse((smk_test$p.value<0.05), print("There exists a trend"),
  print("There exists no trend"))
#Thus there is a trend in the data
#We will split the train and test from the national data to help
  in forecasting
sample_size<-floor(0.8*length(avg_temp))
train_index<-c(1:sample_size)
avg_temp_train<-ts(avg_temp[train_index], frequency = 12)

```

```

avg_temp_test<-ts(avg_temp[-train_index], frequency = 12)
avg_temp_train1<-ts(avg_temp[train_index], frequency = 12)

#We difference the series by a lag of 1 to remove the trend
avg_temp_train<-diff(avg_temp_train)
plot(avg_temp_train, xlab = "Time in Years", ylab = "Differenced
  Temp", main = "Differenced Series(Lag of 1 month)", col = "
  blue")
smk.test(avg_temp_train)
#As the smk test has a higher p-value we can safely say there is
  no trend now in the data
#We now difference by a lag of 12 again to remove the seasonal
  component
avg_temp_train<-diff(avg_temp_train, lag=12)
plot(avg_temp_train, xlab = "Time in Years", ylab = "Differenced
  Temp", main = "Differenced Series(Lag of 12 months after a lag
    of 1 month)", col = "blue")
#As judged by the p-value we can see the series has no upward or
  downward trend thus indicating no trend
#We now employ two tests to check the series is stationary
kpss.test(avg_temp_train)
adf.test(avg_temp_train)
#From the p-values it is clear that the differenced time series
  is now stationary
#We now look at the ACF and PACF plots
par(mfrow=c(1,2))
acf(avg_temp_train, lag.max=120)
pacf(avg_temp_train, lag.max=120)
par(mfrow=c(1,1))
#From the plots it is clear that the series has a seasonal MA(1)
  component. ACF almost cuts off after lag 1 or 2 and PACF
  perseveres indicating a MA(1) or MA(2) process
#We now have two candidate models SARIMA(0,1,1)(0,1,1)[12] and
  SARIMA(0,1,2)(0,1,1)[12]. We fit both of them and see which

```

```

    one is better using RMSE and Ljung-Box Test
#Now we fit a SARIMA(0,1,1)(0,1,1)(12) model
model1 <- arima(avg_temp_train1, order = c(0,1,1),
                seasonal = list(order = c(0,1,1), period = 12))
Box.test(residuals(model1), lag = 12, type = "Ljung")
#Carried out portmanteau test to check whether the residuals are
    iid or not. Low p-value indicates the errors are still
    correlated
#Now we fit a SARIMA(0,1,2)(0,1,1)[12] model
model2 <- arima(avg_temp_train1, order = c(0,1,2),
                seasonal = list(order = c(0,1,1), period = 12))
Box.test(residuals(model2), lag = 12, type = "Ljung")
#High p-value of the model indicates the model has fitted well
#We now also check the RMSE
sqrt(mean(residuals(model1)^2))
sqrt(mean(residuals(model2)^2))
#Both by the RMSE and Ljung-Box test we see that SARIMA(0,1,2)
    (0,1,1)[12] is a better model
#Now we have fit the model. We will forecast for the upcoming 321
    months
h <- length(avg_temp_test)
fc <- predict(model2, n.ahead = h)
fc_ts <- ts(fc$pred,
            start = end(avg_temp_train1) + c(0,1),
            frequency = frequency(avg_temp_train))
avg_temp_test<-ts(avg_temp_test, start = end(avg_temp_train1)+c
    (0,1), frequency = 12)
#Plotting the fitted series
ts.plot(
    avg_temp_test,
    fc_ts,
    col = c("blue", "red"),
    lty = 1,
    xlab = "Time",

```

```
ylab = "Temperature",
main = "Predicted Avg Temperature of India from Dec 1986
      to Aug 2013")
time_forecast <- time(avg_temp_train)[length(avg_temp_train)] +
  (1:h)/frequency(avg_temp_train)
#Calculating the CI for each prediction
upper <- fc$pred + 1.96 * fc$se
lower <- fc$pred - 1.96 * fc$se

# Adding confidence bands
lines(time_forecast, upper, col = "black", lty = 2)
lines(time_forecast, lower, col = "green", lty = 2)
```

THANK YOU